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FrontEnd Assignment

What is Redux

Redux is a state management library used in JavaScript apps especially with React.

It helps you manage and share data (state) across your entire application in a predictable way.

Key Ideas :

Store :- Central place where all app data is kept

Action :- What you want to do (eg. add item, delete item)

Reducer :- Function that updates state based on action

Use Redux

- Keep data organized
- Easy to debug
- Good for large application

Make a react app to make a counter which will start from 0 and 2 buttons with increment and decrement options and count will increase and decrease on the basis of which button.

Basic Objectives :

① Create a Redux - Based Counter App!

- Initial app : 0 (value)
- Two buttons,
 - Increment (+)
 - Decrement (-)
- Count updates based upon button clicks.

* Concepts used :

- Redux toolkit (reduxjs / toolkit)
- React-Redux (useSelector, useDispatch)
- Centralized state management

Steps to make the app :

- Install Dependencies :

```
npm install @reduxjs/toolkit react-redux
```

- Create Redux store :

```
import {ConfigureStore} from 'reduxjs/toolkit';
import counterReducer from '../features/counter/counterSlice';
export const store = ConfigureStore({
  reducer: {
    counter: counterReducer
  }
});
```

- Create counter slice :

```
import {createSlice} from '@reduxjs/toolkit';
const initialState = { value: 0 };
const counterSlice = createSlice({
  name: 'counter',
  initialState,
  reducer: {
    increment: (state) => {
      state.value += 1;
    },
    decrement: (state) => {
      state.value -= 1;
    }
  }
});
```



```
});
```

```
export const { increment, decrement } = CounterSlice.actions;  
export default CounterSlice.reducer;
```

Create Counter Component

```
import React from 'react';  
import { useSelector, useDispatch } from 'react-redux';  
import { increment, decrement } from './counterSlice';  
const Counter = () => {  
  const count = useSelector((data) => state.counter.value);  
  const dispatch = useDispatch();  
  return (  
    <div style={{ textAlign: 'center' }}>  
      <h1> Counter: {count} </h1>  
      <button onClick={() => dispatch(increment())}>  
        </button>  
    </div>  
  );  
};
```

```
};
```

```
export default Counter
```

Connect Redux to App:

```
import React from 'react';  
import ReactDOM from 'react-dom/client';  
import App from './App';  
import { Provider } from 'react-redux';  
ReactDOM.createRoot(document.getElementById('root')).  
  render(<Provider store={store}>  
    <App>  
  );
```


Use Counter in App :

```
import Counter from './location';
```

```
function App() {  
  return (  
    <div>  
      <Counter />  
    </div>  
  );  
};
```

Diagram :

React UI Button Check



Dispatch Action Sent



Reducer Update Count



store Count State



Update UI show
count